EXP NO 20: Implement Future Sales Prediction using a suitable machine learning algorithm

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AIM

To implement Future Sales Prediction using a suitable Machine Learning algorithm and predict future product sales based on historical sales data.

PROCEDURE

1. **Collect the historical sales dataset containing information such as date, product, quantity sold, and sales amount.**
2. **Preprocess the data by handling missing values, converting dates into useful numerical features, and removing unnecessary attributes.**
3. **Split the dataset into training and testing data for model development and evaluation.**
4. **Train a suitable Machine Learning model, such as Linear Regression, using historical sales features to predict future sales.**
5. **Evaluate the model using suitable metrics such as MAE and R^2 score, and use the trained model to predict future sales.**

CODE:

```
# Future Sales Prediction using Linear Regression


import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.linear_model import LinearRegression

from sklearn.metrics import mean_absolute_error, r2_score


# Load dataset from the generated dummy file

df = pd.read_csv("/tmp/sales_data.csv")


…   "Day": [1]
```

```
})

print("Predicted Future Sales:", model.predict(future)[0])
```

OUTPUT:

```
Mean Absolute Error: 32.196935293089766
R2 Score: 0.5205246687013847
Predicted Future Sales: 121.76348291823575
```

Result:

The Future Sales Prediction model was successfully implemented using the Linear Regression algorithm. The model learned from historical sales data and predicted future sales